

Beyond Text and Tables: Vision-Language Model Integration in ComProScanner for Extracting Materials Data from Scientific Figures with High Accuracy

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Summary

ComProScanner is an open-source Python package that automates the extraction of composition-property data from materials science literature, operating as a fully end-to-end pipeline from literature search to structured database construction without human intervention, provided publisher Text and Data Mining (TDM) API keys are supplied. Building on the previously introduced ComProScanner framework (Roy et al., 2026), this release extends it with native vision language model (VLM)-based figure extraction, enabling the recovery of composition-property pairs from scientific charts and plots embedded in journal articles. The package is implemented in Python and orchestrated via the CrewAI framework, giving users the flexibility to configure any LiteLLM-compatible model for both text extraction and figure interpretation.

Statement of need

A substantial fraction of quantitative composition-property data in the materials science literature is reported exclusively in graphical form. For example, bar charts, scatter plots and line graphs encoding piezoelectric coefficients, dielectric constants or mechanical responses across doped compositions are ubiquitous in experimental publications, yet the numerical values they contain are frequently absent from article text or supplementary tables. Existing extraction frameworks, including the prior version of ComProScanner, are limited to textual and tabular content and are therefore systematically blind to this class of information. Specialised plot digitisation tools such as Plot2Spectra (Jiang et al., 2022), MatGD (Lee et al., 2023) and LineEX (P et al., 2023) have demonstrated that automated numerical extraction from figures is feasible, but they operate as standalone digitisers requiring figure-type-specific engineering and are not integrated into any end-to-end literature mining pipeline. The VLM extension presented here addresses this gap by coupling figure extraction natively with ComProScanner's existing publisher API access, multi-agent orchestration and structured output schema, enabling researchers to build composition-property databases that capture data from text, tables and figures within a single automated workflow.

State of the field

Automated extraction of materials data from scientific literature has progressed considerably over the past decade, from rule-based and named entity recognition approaches (Mavracic

et al., 2021; Swain & Cole, 2016) to LLM-based pipelines employing prompt engineering (Dagdelen et al., 2024; Maciej P. Polak & Morgan, 2024) and multi-agent orchestration (Ansari & Moosavi, 2024; Ghosh & Tewari, 2026). However, virtually all existing frameworks operate exclusively on text and tables. The closest multimodal comparator in the materials domain is nanoMINER (Odobesku et al., 2025), which combines YOLO-based visual detection with GPT-4o for nanomaterials information extraction; however, it requires manual article provision and does not offer automated publisher access. On the chart understanding side, models such as DePlot (Liu, Eisenschlos, et al., 2023) and MatCha (Liu, Piccinno, et al., 2023) have demonstrated strong zero-shot performance on general chart comprehension tasks, whilst Zheng et al. (Zheng et al., 2024) and PlotExtract (Maciej P. Polak & Morgan, 2025) have shown that VLMs can extract quantitative data from chemistry and materials figures with high accuracy. Nonetheless, ComProScanner is the first framework to integrate VLM-based figure extraction as a native, configurable component of a fully automated, publisher-integrated, materials-specific composition-property extraction pipeline.

Software design

The VLM extension introduces two principal components into the existing ComProScanner architecture. The first is FigureExtractor, a shared utility that identifies figures whose captions match user-specified keywords and saves them as JPEG images to a DOI-keyed directory, handling JPEG/JPG, as well as transparent PNG and animated GIF formats. The second, but more significant component is GraphExtractorTool, a CrewAI BaseTool that, given a DOI, reads all previously saved figures and submits them to a configurable VLM with a structured prompt designed to recover composition-property value pairs, returning results in the standard ComProScanner JSON schema. The tool is invoked by the Composition-Property Data Extractor agent within the existing five-agent CrewAI workflow. An image-aware fallback has additionally been added to the Materials Data Identifier agent: if the initial text-based RAG step returns a negative result, the agent queries available figures via VLM before deciding whether to proceed with full extraction, preventing articles with graph-only data from being discarded prematurely. The updated overall workflow and the modified multi-agent extraction flow are illustrated in Figure 1. The default VLM is gemini-3-flash-preview, selected on the basis of benchmarking results described in the Research Impact Statement; the vlm_model parameter accepts any LiteLLM-compatible model identifier, allowing users to substitute any VLM of their choice. The release additionally introduces a value_error_thresholds parameter in both evaluate_semantic() and evaluate_agentic(), accepting a dictionary mapping (min, max) range tuples to absolute error tolerances for numeric property value comparisons, institutional token support for off-campus Elsevier access, improved JSON recovery from mixed-text agent outputs and a CalVer versioning scheme (YYYY.MM.DD).

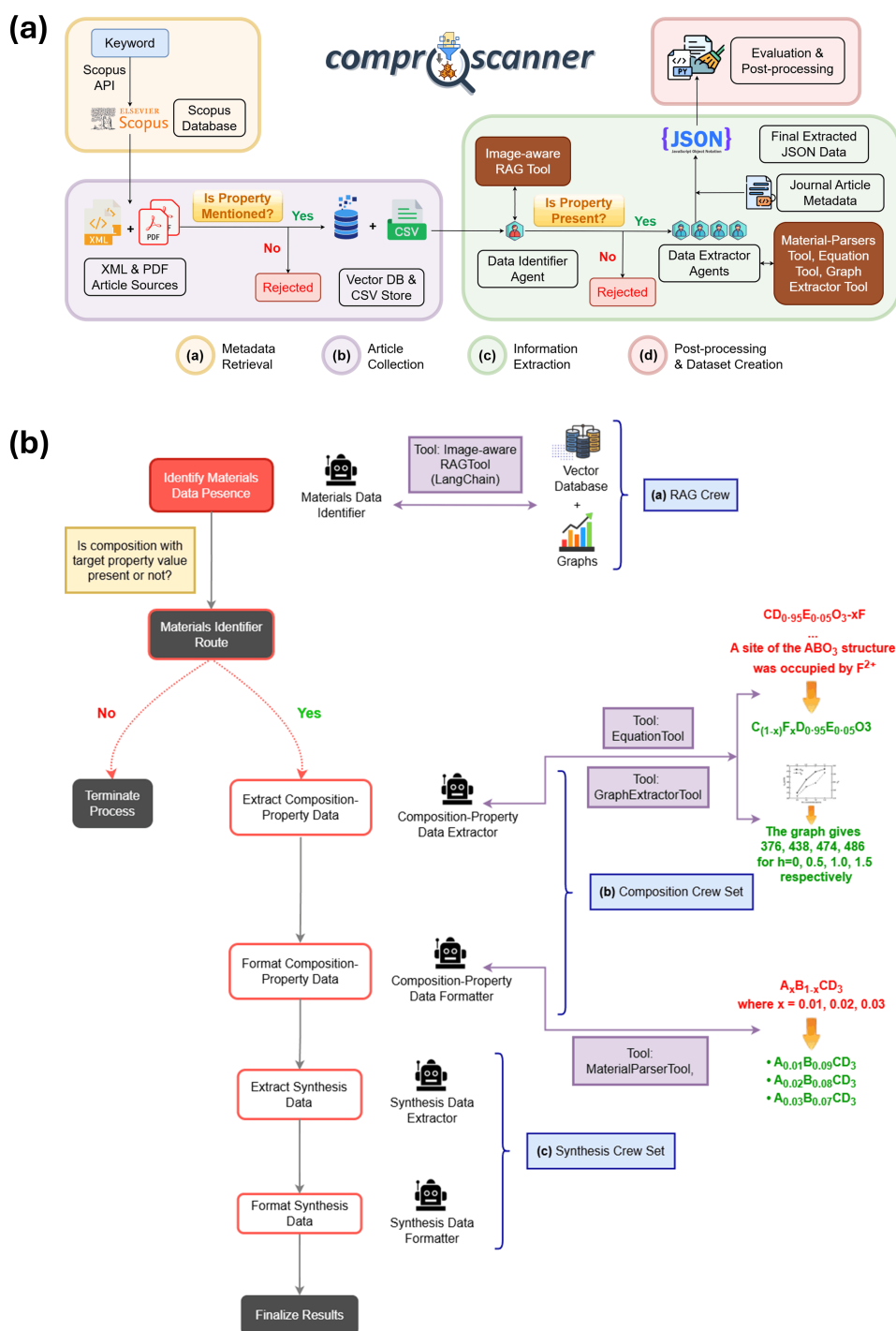


Figure 1: (a) Overall workflow diagram of ComProScanner framework incorporating the GraphExtractorTool and EquationTool. (b) Flow diagram of ComProScanner framework's information extraction process incorporating the image-aware RAGTool, GraphExtractorTool, EquationTool and Material-ParserTool. The detailed descriptions for other components can be found in the original ComProScanner paper (Roy et al., 2026).

Research impact statement

To evaluate the capability of cost-effective VLMs for composition-property data extraction from scientific figures, four VLMs were benchmarked on 50 piezoelectric ceramic articles from the established d_{33} test corpus described in our previous paper (Roy et al., 2026). Models were selected from the LMArena VLM Leaderboard (Diagram category) (Chiang et al., 2024) using simultaneous criteria of an Arena ELO score of at least 1,250 and an input cost below \$1.50 per million tokens, yielding Gemini-3-Flash-Preview (Pichai et al., 2025), Gemini-2.5-Pro (Comanici et al., 2025), GPT-5-Chat-Latest (Singh et al., 2026) and GPT-5.1 (OpenAI, 2026). Evaluation was performed on the composition_property_values field using the ComProScanner semantic evaluator with range-based value error thresholds of ± 0.5 , ± 1 , and ± 2 pC/N applied across different value ranges. Gemini-3-Flash-Preview achieved the highest performance across all metrics, with a composition accuracy of 0.97 and a normalised F1 score of 0.97, whilst simultaneously commanding the lowest input cost among the evaluated models. These results, summarised in Figure 2, motivate its adoption as the default VLM for the GraphExtractorTool. The ComProScanner package is associated with well tested functionality using pytest and comprehensive API documentation at <https://slimeslab.github.io/ComProScanner> under the MIT License.

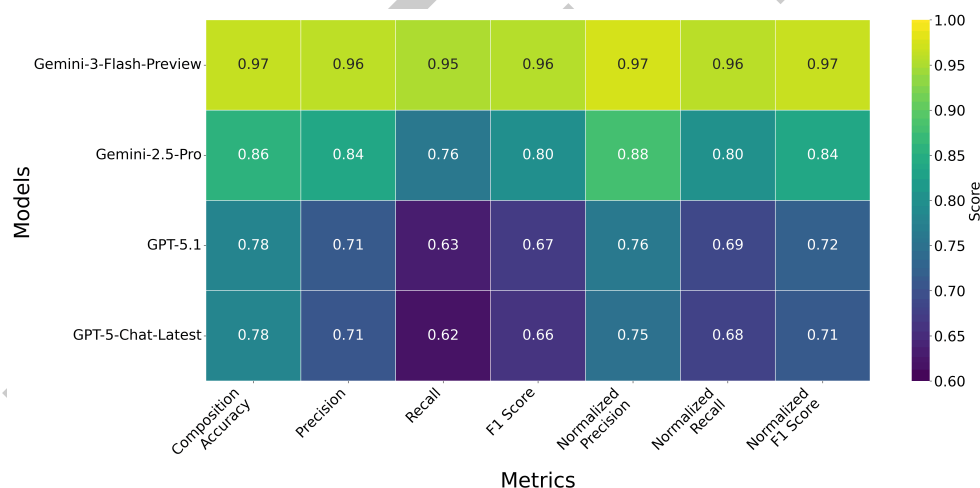


Figure 2: Confusion matrix from semantic evaluation with 1.0 threshold for composition-property data, showcasing all 7 evaluation parameters, such as weight-based composition accuracy, classification metrics (precision, recall and F1-score) and normalised classification metrics (normalised precision, normalised recall and normalised F1-score), across 4 different VLMs used in this study.

AI Usage Disclosure

The core software architecture was designed by the authors without the use of generative AI. However, Claude Code (Sonnet 4.6) has been used for refactoring, debugging and limited generation of boilerplate code, some tests and documentation. No AI generated content has been accepted without review and testing, nor does it constitute the majority of the work produced. Finally, Claude Sonnet 4.6 was used to suggest edits for clarity and structure, but was not used to generate the contents or sections while writing the paper.

Author Contributions

AR: conceptualisation, software, data curation, formal analysis, investigation, methodology, validation, writing - original draft. EG: resources, supervision. JB: conceptualisation, formal

analysis, funding acquisition, investigation, resources, validation, writing - review & editing, supervision. CG: conceptualisation, formal analysis, funding acquisition, investigation, resources, validation, writing - review & editing, supervision.

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References

- Ansari, M., & Moosavi, S. M. (2024). Agent-based learning of materials datasets from the scientific literature. *Digital Discovery*, 3(12), 2607–2617. <https://doi.org/10.1039/D4DD000252K>
- Chiang, W. L., Zheng, L., Sheng, Y., Angelopoulos, A. N., Li, T., Li, D., Zhang, H., Zhu, B., Jordan, M., Gonzalez, J. E., & Stoica, I. (2024). *Chatbot arena: An open platform for evaluating LLMs by human preference*. <https://doi.org/10.48550/arXiv.2403.04132>
- Comanici, G., Bieber, E., Schaeckermann, M., Pasupat, I., Sachdeva, N., Dhillon, I., Blistein, M., Ram, O., Zhang, D., Rosen, E., Marris, L., Petulla, S., Gaffney, C., Aharoni, A., Lintz, N., Pais, T. C., Jacobsson, H., Szpektor, I., Jiang, N.-J., ... Helmholtz, W. (2025). *Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities*. <https://doi.org/10.48550/arXiv.2507.06261>
- Dagdelen, J., Dunn, A., Lee, S., Walker, N., Rosen, A. S., Ceder, G., Persson, K. A., & Jain, A. (2024). Structured information extraction from scientific text with large language models. *Nature Communications*, 15(1), 1418. <https://doi.org/10.1038/s41467-024-45563-x>
- Ghosh, S., & Tewari, A. (2026). LLM-based AI agents for automated extraction of material properties and structural features. *Computational Materials Science*, 265, 114521. <https://doi.org/10.1016/j.commatsci.2026.114521>
- Jiang, W., Li, K., Spreadbury, T., Schwenker, E., Cossairt, O., & Chan, M. K. (2022). Plot2Spectra: An automatic spectra extraction tool. *Digital Discovery*, 1(5), 719–731. <https://doi.org/10.1039/D1DD00036E>
- Lee, J., Lee, W., & Kim, J. (2023). MatGD: Materials graph digitizer. *ACS Applied Materials & Interfaces*, 16(1), 723–730. <https://doi.org/10.1021/acsami.3c14781>
- Liu, F., Eisenschlos, J., Piccinno, F., Krichene, S., Pang, C., Lee, K., Joshi, M., Chen, W., Collier, N., & Altun, Y. (2023). DePlot: One-shot visual language reasoning by plot-to-table translation. In A. Rogers, J. Boyd-Graber, & N. Okazaki (Eds.), *Findings of the association for computational linguistics: ACL 2023* (pp. 10381–10399). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.findings-acl.660>
- Liu, F., Piccinno, F., Krichene, S., Pang, C., Lee, K., Joshi, M., Altun, Y., Collier, N., & Eisenschlos, J. M. (2023). *MatCha: Enhancing visual language pretraining with math reasoning and chart derendering*. <https://doi.org/10.48550/arXiv.2212.09662>
- Mavracic, J., Court, C. J., Isazawa, T., Elliott, S. R., & Cole, J. M. (2021). ChemDataExtractor 2.0: Autopopulated ontologies for materials science. *Journal of Chemical Information and Modeling*, 61(9), 4280–4289. <https://doi.org/10.1021/acs.jcim.1c00446>
- Odobesku, R., Romanova, K., Mirzaeva, S., Zagorulko, O., Sim, R., Khakimullin, R., Razlivina, J., Dmitrenko, A., & Vinogradov, V. (2025). Agent-based multimodal information extraction

- 147 for nanomaterials. *Npj Computational Materials*, 11(1), 194. <https://doi.org/10.1038/s41524-025-01674-7>
- 148
- 149 OpenAI. (2026). *GPT-5.1: A Smarter, More Conversational ChatGPT*. <https://openai.com/index/gpt-5-1/>.
- 150
- 151 P, S. V., Yusuf Hassan, M., & Singh, M. (2023). LineEX: Data extraction from scientific line
- 152 charts. *2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*,
- 153 6202–6210. <https://doi.org/10.1109/WACV56688.2023.00615>
- 154 Pichai, S., Hassabis, D., & Kavukcuoglu, K. (2025). *A new era of intelligence with gemini 3*.
- 155 <https://blog.google/products-and-platforms/products/gemini/gemini-3/>.
- 156 Polak, Maciej P., & Morgan, D. (2024). Extracting accurate materials data from research papers
- 157 with conversational language models and prompt engineering. *Nature Communications*,
- 158 15(1), 1569. <https://doi.org/10.1038/s41467-024-45914-8>
- 159 Polak, Maciej P., & Morgan, D. (2025). *Leveraging vision capabilities of multimodal LLMs for*
- 160 *automated data extraction from plots*. <https://doi.org/10.48550/arXiv.2503.12326>
- 161 Roy, A., Grisan, E., Buckeridge, J., & Gattinoni, C. (2026). ComProScanner: A multi-
- 162 agent based framework for composition-property structured data extraction from scientific
- 163 literature. *Digital Discovery*, 5(4), 1794–1808. <https://doi.org/10.1039/D5DD00521C>
- 164 Singh, A., Fry, A., Perelman, A., Tart, A., Ganesh, A., El-Kishky, A., McLaughlin, A., Low,
- 165 A., Ostrow, A., Ananthram, A., Nathan, A., Luo, A., Helyar, A., Madry, A., Efremov, A.,
- 166 Spyra, A., Baker-Whitcomb, A., Beutel, A., Karpenko, A., ... Wang, Z. (2026). *OpenAI*
- 167 *GPT-5 system card*. <https://doi.org/10.48550/arXiv.2601.03267>
- 168 Swain, M. C., & Cole, J. M. (2016). ChemDataExtractor: A toolkit for automated extraction
- 169 of chemical information from the scientific literature. *Journal of Chemical Information and*
- 170 *Modeling*, 56(10), 1894–1904. <https://doi.org/10.1021/acs.jcim.6b00207>
- 171 Zheng, Z., He, Z., Khattab, O., Rampal, N., Zaharia, M. A., Borgs, C., Chayes, J. T., &
- 172 Yaghi, O. M. (2024). Image and data mining in reticular chemistry powered by GPT-4V.
- 173 *Digital Discovery*, 3(3), 491–501. <https://doi.org/10.1039/D3DD00239J>